

CEAS Co-op Reflection System

D0 High-Level Design

Team Members:

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1. Title, Goal Statement, and Conventions

Title: Design D0: High-Level System Architecture

Goal Statement: Let CEAS co-op students reflect on each completed rotation using rotation-specific questions, give assigned advisors a clear summary with workplace supervisor observations, and keep each reflection private to the student and their assigned advisor.

Basic Input: Student reflections, supervisor evaluations, and university records with rotation numbers and advisor assignments.

Basic Output: Rotation-specific prompts and private reflection summaries with supervisor observations for students and assigned advisors.

Diagram Conventions:

- Solid rectangles: Major components developed by the team.
- Dashed rectangles: External systems or third-party services.
- Cylinders: Databases or persistent data storage.
- Directed arrows: The direction of information flow between components.
- Interface labels (I1, I2, ...): Connections corresponding to entries in the interface specification table.
- Data-flow arrow labels: The information transferred and its form at that stage, such as "validated record (JSON)."

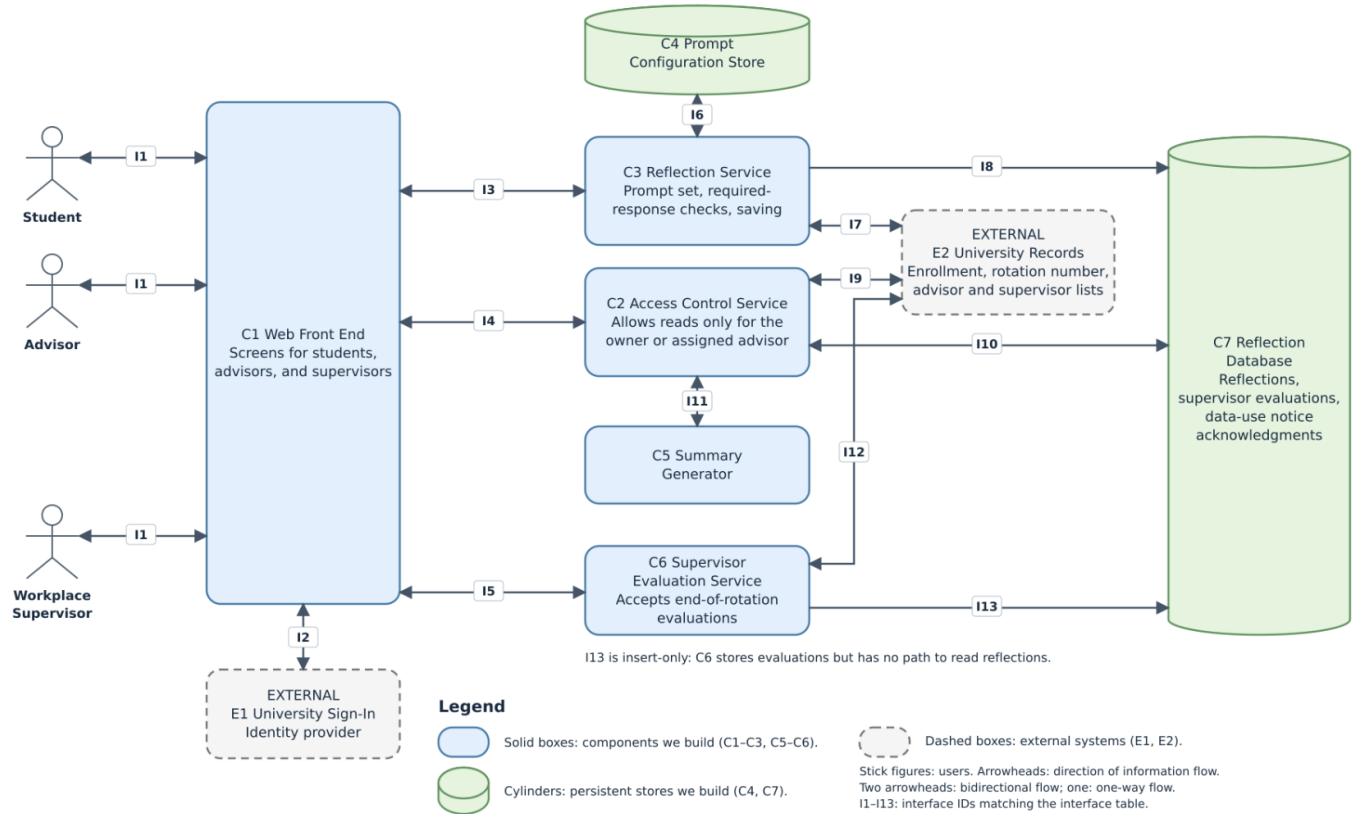
2. Block Diagram (D0 Diagram)

CEAS Co-op Reflection System | D0 High-Level Design: Block Diagram

Goal: Let CEAS co-op students reflect on each completed rotation using rotation-specific questions, give assigned advisors a clear summary with workplace supervisor observations, and keep each reflection private to the student and their assigned advisor.

Input: Student reflections, supervisor evaluations, and university records with rotation numbers and advisor assignments.

Output: Rotation-specific prompts and private reflection summaries with supervisor observations for students and assigned advisors.



3. Component Responsibility Table

Component	Responsibility	Interfaces In	Interfaces Out	Primary Owner
C1 Web Front End	Provides the user-facing screens used by students, advisors, and workplace supervisors to interact with the reflection system.	I1, I2, I3, I4, I5	I1, I2, I3, I4, I5	Maria Malik
C2 Access Control Service	Determines whether a user is allowed to access reflection information based on their role and assigned relationship.	I4, I8, I10, I11	I4, I8, I10, I11	Hinna Parwez
C3 Reflection Service	Provides the appropriate reflection prompts, checks required responses, and saves submitted reflections.	I3, I6, I7	I3, I6, I7, I9	Hinna Parwez
C4 Prompt Configuration Store	Stores the reflection prompts and configuration used for different co-op rotations.	I6	I6	Priyanshi Kulshrestha
C5 Summary Generator	Generates a summary of reflection information that can be shown to an authorized advisor.	I11	I11	Mbathio Lo
C6 Supervisor Evaluation Service	Accepts workplace supervisor evaluations at the end of a co-op rotation.	I5, I12	I5, I13	Aradana Nair
C7 Reflection Database	Stores reflection records, supervisor evaluations, acknowledgment information, and related data used by the system.	I9, I10, I13	I10	Priyanshi Kulshrestha

4. Interface Specification Table

ID	From	To	Input/Output Data	Format	Protocol	Error Handling
I1	User	C1 Web Front End	Input: page requests, form entries, and user actions. Output: pages, validation messages, and confirmations.	HTML / form data	HTTPS	C1 displays an error message and keeps the user's entered information when possible.
I2	C1 Web Front End	E1 University Sign-In	Input: sign-in request. Output: authenticated user information.	Authentication Data	HTTPS	If sign-in fails, C1 asks the user to try signing in again and does not allow access until authentication succeeds.
I3	C1 Web Front End	C3 Reflection Service	Input: reflection responses, placement information, and submission action. Output: prompts, validation results, and submission status.	JSON	HTTPS REST	C3 identifies missing required responses. C1 displays the error and keeps the student's entered responses.
I4	C1 Web Front End	C2 Access Control Service	Input: user identity and requested reflection. Output: reflection information or access denied.	JSON	HTTPS REST	If the user is not authorized, C2 denies access and no reflection information is returned.
I5	C1 Web Front End	C6 Supervisor Evaluation Service	Input: student/placement information and supervisor evaluation. Output: submission status or confirmation.	JSON	HTTPS REST	Missing information is identified and the evaluation is not submitted until the required fields are completed.
I6	C3 Reflection Service	C4 Prompt Configuration Store	Input: rotation number. Output: the prompt set for that rotation.	Configuration Data	Internal Request	If the rotation is unsupported or prompts cannot be found, C3 displays an error instead of showing an incomplete prompt set.
I7	C3 Reflection Service	E2 University Records Source	Input: student and placement information. Output: eligibility, rotation number, and placement information.	JSON	HTTPS API	If a rotation number is unavailable, the student can provide it. If eligibility cannot be confirmed, the

						reflection cannot be started.
I8	C2 Access Control Service	E2 University Records Source	Input: advisor and student information. Output: advisor assignment status.	JSON	HTTPS API	If the advisor assignment cannot be confirmed, access to the reflection is denied.
I9	C3 Reflection Service	C7 Reflection Database	Input: completed reflection record. Output: saved reflection ID/status.	Database Record	Database Connection	If the reflection cannot be saved, no incomplete record is stored and C1 tells the student to try again.
I10	C2 Access Control Service	C7 Reflection Database	Input: authorized reflection request. Output: stored reflection and related evaluation information.	Database Record	Database Connection	If the requested record cannot be found or the database is unavailable, no reflection information is returned.
I11	C2 Access Control Service	C5 Summary Generator	Input: authorized reflection content. Output: reflection summary.	JSON	Internal Request	If a summary cannot be generated, the original reflection can still be shown to the authorized user.
I12	C6 Supervisor or Evaluation Service	E2 University Records Source	Input: supervisor, student, and placement information. Output: supervisor assignment status.	JSON	HTTPS API	If the supervisor's assignment cannot be confirmed, the evaluation is not accepted.
I13	C6 Supervisor or Evaluation Service	C7 Reflection Database	Input: completed supervisor evaluation. Output: saved evaluation ID/status.	Database Record	Database Connection	If the evaluation cannot be saved, no incomplete record is stored and the supervisor is asked to try again.

Example Payload for I3 – Reflection Submission (JSON):

```
{
  "action": "submit",
  "placement_id": "P1024",
  "rotation_number": 2,
  "responses": [
```

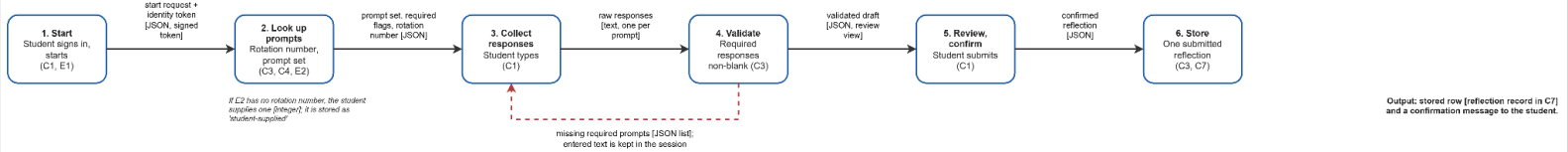
```
{
  "prompt_id": "Q1",
  "text": "I improved my communication skills while working with my team."
},
{
  "prompt_id": "Q2",
  "text": "One challenge was learning how to use unfamiliar software."
}
]
}
```

5. Data-Flow Diagram

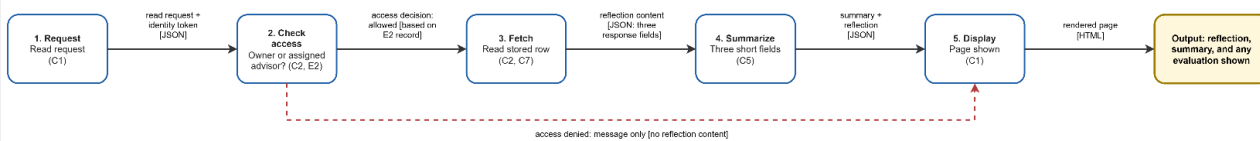
CEAS Co-op Reflection System | D0 High-Level Design: Data-Flow Diagram

Goal: Let CEAS co-op students reflect on each completed rotation with questions that match their rotation number, give assigned advisors a clear summary plus workplace supervisor observations, and keep each reflection private to the student and their assigned advisor.

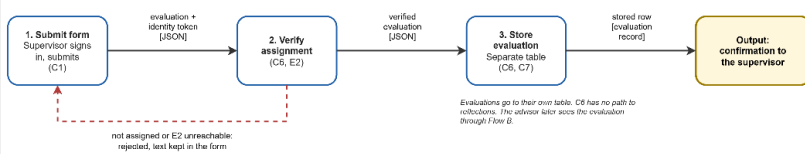
Flow A: A student completes a rotation reflection (US-01, UC-01)



Flow B: An advisor (or the student) reads a reflection and its summary (US-02, US-04, UC-02)



Flow C: A workplace supervisor submits an end-of-rotation evaluation (US-03)



Timing budget

The user stories set no timing requirement. Working target, to be confirmed by the team: each page load or submit finishes in under 2 seconds under normal load. Nothing in this system is real-time or hardware timed.

Legend

- Stage (components in brackets)
- Output the person receives

- Solid arrow: data moving between stages, written as name [form at that point]
- Dashed red arrow: error or denial path

6. Architecture Pattern and Justification

Chosen Pattern(s): The CEAS Co-op Reflection System primarily uses client-server architecture. The C1 Web Front End acts as the client, while the backend components like the Reflection Service, Access Control Service, Summary Generator, and Supervisor Evaluation Service act as server-side services. The system follows a layered architecture, separating the user interface, application logic, access control, and data storage. The pipeline pattern is used for the reflection workflow as data moves from the student interface through validation and processing before being stored in the Reflection Database.

Fit to the Problem: Client-server fits the project because students, advisors, and workplace supervisors need to access the same centralized system through a web interface. Separating the interface, business logic, and database also supports the different responsibilities already defined for the system components.

Team Skills: The architecture matches the team's experience with web development, backend services, databases, and APIs. It also lets team members work on different parts of the system while keeping the connections between components clear.

Performance and Timing: The system does not need instant real-time processing. The reflection submissions, summaries, and supervisor evaluations can be handled through normal web requests. Tasks that are slower, like generating summaries, can be handled separately from the front end.

Scalability: The design can support more students, advisors, reflections, and co-op rotations as the system grows. Backend services can also be expanded without changing the whole system.

Hardware Constraints: No special hardware. Users bring an ordinary browser on a laptop or phone.

Rejected Pattern: The team rejected microservices as the main architecture. Microservices separates the system into many independently deployed services, however that would add extra complexity for a project of this size. The client-server and layered design is simpler while still keeping the system organized.

7. Decision Log

Decision	Alternatives Considered	Reason
Use university sign-in for user authentication.	Create our own account and password system.	University sign-in avoids having to manage passwords ourselves and aligns with existing university records.
Keep reflection prompts in a separate Prompt Configuration Store.	Hard-code the prompts directly into the Reflection Service.	Keeping the prompts separate makes them easier to update if questions or rotation requirements change.
Use one Access Control Service to manage access to reflections.	Have each service check access separately.	A single Access Control Service keeps access rules consistent and helps make sure reflections are only available to the student and their assigned advisor.
Keep supervisor evaluations separate from student reflections.	Handle reflections and supervisor evaluations through the same service.	Keeping them separate helps prevent workplace supervisors from gaining access to private student reflections.
Allow students to enter their rotation number if it is unavailable from university records.	Prevent the student from continuing until the university record is available.	Allowing the student to enter the rotation number lets them continue their reflection even if that information is temporarily unavailable.